CS612 Lecture 03 — In-Class Exercises

Randomized Algorithms & Probabilistic Analysis · Sat 10 Oct 2026 · ungraded

Opening

(1) What would it mean for a sorting algorithm to be "safe" against an adversary who crafts inputs? (2) What would it mean to "check" a claimed matrix product $C = AB$ in $O(n^2)$ time? In each case, what are we willing to give up?

A — Classify the guarantee

For each algorithm: Las Vegas or Monte Carlo? One- or two-sided error? What is random and what is bounded?

1. Randomized quicksort
2. Freivalds' algorithm
3. Miller–Rabin primality test
4. Rabin–Karp that reports a match whenever the hashes are equal
5. Rabin–Karp that verifies every hash match character by character
6. Estimating the median as the median of a random sample of 1000 values

B — Analysis practice

1. In randomized quicksort, compute the probability that z_1 and z_n are compared, and the probability that z_i and z_{i+1} are compared. Explain both intuitively.
2. Using the lemma $\Pr[z_i, z_j \text{ compared}] = 2/(j - i + 1)$, show that $\mathbb{E}[X] < 2nH_n$.
3. (Harder) Prove the three-group bound $\mathbb{E}[\text{comparisons}] \le 4n + O(1)$ for randomized quickselect.

C — Design with randomness

1. **Budget.** For $n = 10^4$, how many Freivalds trials give error $\le 10^{-12}$? Compare the cost with recomputing AB.
2. **LV → MC.** Randomized quickselect uses $\le 4n$ comparisons in expectation. Give a Monte Carlo version that always stops within $40n$ comparisons, and bound its failure probability. Can you get failure probability 2^{-10} within $100n$ comparisons?
3. **Reservoir sampling ($k = 1$).** Keep $s = x_1$; for $t \ge 2$ replace s by x_t with probability $1/t$. Prove that after n items $\Pr[s = x_i] = 1/n$ for every i. Generalise to a sample of size k.
4. **Adversary.** A web service sorts user-supplied arrays with randomized quicksort, seeded with the server start time; response times are visible. Describe an attack, and a fix.

Take-home (optional)

Run `Student_Resources/freivalds.cpp`. Modify it to use $r \in \{1, \dots, 100\}^n$ and measure the single-trial detection rate for a single wrong entry. What does theory predict?